

Tornado: A Python Web Framework That's Simple and Quick



This article presents Tornado, the Python Web framework. Originally developed at FriendFeed, Tornado is an asynchronous networking library. It is able to scale up to tens of thousands of open connections with the help of its non-blocking network I/O.

A part from introducing you to Tornado, this article explores the extent to which it can be scaled, its features, the installation process, and its advantages compared to other Python Web frameworks. This article is meant for those interested in knowing more about Web designing as well as Web developers. It may even be helpful to serious Python developers or contributors trying to explore ways to improve their Web development skills.

A framework is a collection of programs that uses a control mechanism to structure code to build a bigger system. We can develop simple websites using basic designing languages like HTML, CSS, Python, etc. However, considering the complex and dynamic nature of the Web, it is difficult to do this at a low cost and within the short deadlines that sites are typically developed. Hence, developers want something that empowers them and has fewer bugs. The purpose of a Web framework is to provide some features that are essential for all kinds of applications, such as database support, URL handling, HTML templating, etc.

Python is one of the most used general-purpose programming languages. It is also used as a scripting language but mostly in non-scripting contexts. Python's environment consists of a number of Web frameworks, ranging from the minimalistic to the fully bloated. Tornado is one of the best Web frameworks of the Python programming language. It

is Facebook's real time Python Web framework (we will go into this later). It enables a clean approach to Web server programming, focuses on non-blocking operations and scales to an enormous number of open connections. Tornado is an asynchronous networking library that was originally developed at FriendFeed, a kind of social network that enables you to share the issues you found on the Web and it allows discussions. FriendFeed was acquired by Facebook in 2009 and Tornado was open sourced soon after. Django, CherryPy, web2Py, Grok, Pylons and TurboGears are some of the other frameworks.

Why Tornado?

So why is Tornado preferred, even though Python has a number of frameworks? Well, Tornado is noted for its high performance and supposedly has the capacity to solve the C10k problem (problem of optimising network related issues so that it can handle a large number of clients at the same time) that affects other servers. Tornado focuses on speed and handling large volumes of simultaneous traffic.

Let's look at those features of Tornado that differentiate it from other frameworks.

Built-in support: Tornado comes with built-in support and finds solutions for most of the tedious aspects of Web development like templates, localisation, signed cookies, etc. Tornado also allows users to mix it with other frameworks,